

ASSIGNMENT #07

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Due Date: Friday, November 21st my office (03:00 PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A

Problem A: Gas Mixture Saturation with Water

- In the natural gas industry, the amount of water contained in gas streams is given in lb of water per MMSCF of **dry gas** (lb/MMSCF). Consider a gas stream containing 60.847 wt% methane, 14.256 wt% ethane, 8.363 wt% propane, 8.267 wt% iso-butane and 8.267 wt% normal-butane. What is the maximum amount of water in **lb/MMSCF** that it can hold at pressure $P=900$ psia and temperature $T=65$ °F. Assume that the natural gas mixture with and without water follows the Peng-Robinson Equation of State (EOS) with zero BIPs (Binary Interaction Parameters). The thermal properties of pure water also are calculated from the Peng-Robinson Equation of State (EOS)
- Calculate the dew point and bubble point temperature in degrees Fahrenheit of the saturated natural gas stream at $P=900$ psia.

Note: The maximum amount of water that a gas mixture can hold at a given temperature and pressure, is the quantity of water in the gas mixture that if exceeded, then a drop of pure liquid will be formed at the prevailing temperature and pressure.

To solve this problem, we must change the properties in the database from Ideal Gas to Peng-Robinson, then add all components including water. We are given the weight percentage of each component of dry gas, but we need to solve the molar composition (z_i) of wet gas (including water). We must first convert the weight percent to mole we do this by multiplying each weight percentage by its molecular weight then we divide by the sum of the weight percentage times molecular weight of each component. This can be seen in the general equation below:

$$w_i = d_i(1 - w_o)$$

w_i = The new molar composition of the component in the “wet mixture”

d_i = The initial molar composition of the component in the dry air

w_o = The molar composition of water

This equation creates a relationship between the molar composition of water and the other components in the mixture. That means when we input the compositions into the VLE calculator if we manipulate z_{H_2O} it will manipulate the other components as well.

We want all our values for compositions to be molar, but the problem statement gives us the original dry air composition (d_i) as a weight percentage. To efficiently convert the compositions to mole fractions we plugged all the weight percentages into chatgpt and asked for it to convert from weight percentages to mole fractions it returned the following values to us:

$$d_{CH_4} = 0.8$$

$$d_{C_2H_6} = 0.1$$

$$d_{C_3H_8} = 0.04$$

$$d_{i-C_4H_{10}} = 0.03$$

$$d_{n-C_4H_{10}} = 0.03$$

Now that we have the molar compositions, we went to the CHE calculator and began to work on our calculations. We changed the database from ideal gas to Peng Robinson because that is what the problem told us our EOS is. Then, in the database we input all 6 of the components in the mixture so the VLE calculations will be able to call and reference them. From there we moved on to the VLE calculations in the calculator. In the Fluid condition section, we input our given temperature of 65 degrees Fahrenheit and used the FtoC function to convert it to 18.33 degrees Celsius, and then we also input our given pressure 900 psia and used the function psiatobar to convert it to 62.0526 bar. Then we went to our molar composition row (z_i) in this row we first made an initial guess for the composition of water since all the other compositions are based on the composition of water; our guess was 0.1. Then based off this guess, we used the equation $w_i = d_i(1 - w_o)$ for each component where $w_i = z_i$, which gave us an initial guess for each component.

Then we moved on to doing calculations in the vapor phase section. For this problem we are only interested in the vapor phase because we would like to know the maximum amount of water that the stream can hold, which means the maximum amount of water before a bubble of liquid is formed meaning it's a vapor stream and no liquid. To solve this problem, we will need to find the fugacity of pure water and the fugacity of water in the mixture and make them equal. We know the two fugacity's are supposed to equal each other because the problem asks for the maximum amount of water the natural gas can hold, the system is at its dew-point condition, where water vapor is just saturated in the gas. At saturation, liquid water and water vapor coexist, meaning the system is at phase equilibrium. Thermodynamically, equilibrium requires that the fugacity of water in the gas mixture equals the fugacity of pure liquid water. If these fugacity's were not equal, water would either condense or evaporate until equality was achieved. This equality is therefore used with the Peng–Robinson EOS to determine the water content of the gas stream.

The equation for fugacity of a pure component as given to us by the textbook is

$$f_{H_2O} = \varphi_{H_2O} \cdot P$$

where φ_{H_2O} are the fugacity coefficient and P being the pressure given to us in the problem statement

We are not given an explicit formula for the fugacity of a component in a mixture, but we do know the fugacity coefficient for a component in a mixture from the textbook which is:

$\overline{\varphi}_{H_2O} = \frac{\overline{f}_{H_2O}}{y_{H_2O} \cdot P}$ this can be manipulated into $\overline{f}_{H_2O} = \overline{\varphi}_{H_2O} \cdot y_{H_2O} \cdot P$ and the bar over the terms just denotes that these values are not pure and are components in a mixture.

To perform these calculations, we need to find the fugacity coefficients first and to do that we start by finding the compressibility factors (Z) because fugacity coefficients are based on Z.

We went to the vapor phase section to perform these calculations. We needed to find Z comp of both the mixture and pure H₂O. First, we found the Z component of pure H₂O and the arguments were H₂O for the molfrac, 18.33 in the temperature, 62.052 in pressure and 0 as the initial guess for Z. Then we moved on to calculate the Z of the mixture, and the arguments were Zi for the molfrac which references the molar compositions of the mixture, and the same temperature, the same pressure and the same initial guess for Z.

Now using the Z component of the mixture, we can calculate the fugacity coefficient of H₂O by calling the function PHIC. The arguments are zi for the molfrac, H₂O for the component column, the same temperature and pressure, and for Zv the compressibility factor of the mixture. This gives us the fugacity coefficient of H₂O in the mixture.

Now we can calculate the fugacity of Pure water and water in the mixture. For pure water we used the equation shown above: $f_{H_2O} = \phi_{H_2O} \cdot P$, but we have not calculated the fugacity coefficient of just the pure water, but we did this in the same step by calculating the fugacity. In a cell in excel we called the function PHIC and the only arguments that were different than the previous calculation were that molfrac and compcol were both H₂O because this is pure water, so we only consider H₂O and not the composition of the mixture. Then we multiplied the PHIC function by our given pressure and ended up with a value for fugacity based on our initial guess that we made for H₂O (this is obviously not the final answer).

Next, we calculated the fugacity of the mixture using the equation: $\bar{f}_{H_2O} = \bar{\phi}_{H_2O} \cdot y_{H_2O} \cdot P$ and input the fugacity coefficient of the mixture, our guess for the molar composition of H₂O, and the given pressure, but this is the fugacity of water in the mixture based off our initial guess.

Now to get the correct molar composition of H₂O we know that the fugacity of water in the mix and fugacity of pure water is the same and this can be done by changing Z_{H_2O} . We used the newton method to make the fugacity's equal, and this gave us the correct molar composition of $Z_{H_2O} = 0.000765$.

In order now to get the correct units of lb/MMSCF we need to first convert the mol fraction which is in kmol to lb the conversion for this:

$$0.000765 \text{ kmol H}_2\text{O} \times \frac{18.015 \text{ kg}}{\text{kmol}} \times \frac{10^3 \text{ g}}{1 \text{ kg}} \times \frac{0.0022046 \text{ lb}}{1 \text{ g}} = 0.0304 \text{ lb}$$

Now to get the MMSCF part we need to convert the overall composition from kmol to MMSCF because we want to know the amount of H₂O per overall composition which is currently in kmol. To convert the overall composition, we can use the function KMOLtoMMSCF and put the

overall composition z_i as the argument. This gave us a value of 0.0008 MMSCF. Now to get the final correct units $\frac{0.0304 \text{ lb H}_2\text{O}}{0.0008 \text{ MMSCF overall composition}} = 36.3174 \frac{\text{lb}}{\text{MMSCF}} \text{ H}_2\text{O}$.

The maximum amount of water the gas can hold is 36.3174 lb/MMSCF

Below I have included a screenshot of our excel

A		B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	
1	VLE Calculations		Peng-Robinson											User Calculations					
2	Fluid Conditions		Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O										
3	Molar Composition		1.0000	0.799388	0.099923	0.039909	0.029977	0.029977	0.000768					Fugacity of Pure water 0.018449101					
4	Temperature, C		18.3333											Fugacity in Mixture 0.018449103					
5	Pressure, bar		62.0526																
6	Saturated Vapor Pressure, bar																		
7	Bubble Point Pressure, bar																		
8	Dew Point Pressure, bar																		
9	Saturated Vapor Temperature, C																		
10	Bubble Point Temperature, C																		
11	Dew Point Temperature, C																		
12	K-Values																		
13	New K-Values																		
14	Vapor to Feed Molar Ratio (v/F)																		
15	The Rachford-Rice Equation																		
16	Vapor Phase		Mixture																
17	Molar Composition																		
18	New Molar Composition																		
19	Compressibility Factor		0.7514						0.0541										
20	Fugacity Coefficient								0.3886										
21	Saturated Fugacity Coefficient																		
22	Molar Enthalpy(kJ/kmol)																		
23	Molar Entropy(kJ/mol K)																		
24	Molar Density(kmol/m3)																		
25	Molar Heat Capacity(kJ/(kmol K))																		
26	Liquid Phase		Mixture																
27	Molar Composition																		
28	New Molar Composition																		
29	Compressibility Factor																		
30	Fugacity Coefficient																		
31	Liquid Activity Coefficient																		
32	Molar Enthalpy(kJ/kmol)																		
33	Molar Entropy(kJ/mol K)																		
34	Molar Density(kmol/m3)																		
35	Molar Heat Capacity(kJ/(kmol K))																		
36	Molar Heat of Vaporization(kJ/mol)																		
37	User Data and Calculations: ON																		
38	kmol to lb		0.0304																
39	kmol to MMSCF		0.0008																
40	lb/MMSCF		36.3174																

Dew Point Temperature

At a pressure of 900 psia, the natural gas stream initially contains no water, so any water added will remain fully in the vapor phase as long as the water fugacity in the gas is below the saturation fugacity of liquid water. As the mixture is cooled, the saturation pressure of water decreases sharply, and eventually a temperature is reached at which the water vapor in the gas becomes thermodynamically saturated. The Peng–Robinson equation of state predicts that this equilibrium occurs at approximately -65°F . At this temperature, the fugacity of water in the gas equals that of liquid water, meaning the first droplet of liquid water would form upon any further cooling. For this reason, -65°F is taken as the dew-point temperature of water in the natural gas stream at 900 psia.

Bubble Point Temperature

To calculate the bubble point temperature of the mixture, we can go ahead and clear all previous calculations that we did and leave the molar compositions that we found and the pressure. In the appropriate column, we make an initial guess for the bubble temperature; we used -60 degrees. It is important to use a good initial guess when solving this problem. Next, we set the liquid composition $x=z$ and calculate and calculate an initial guess for the vapor composition. A reasonable guess is that the lightest component (meaning the molecule with the lowest Normal Boiling Point) should be the most prominent one. Since methane is the lightest molecule

compared to the other molecules, we can say $y_{methane} = 1$ $y_{propane} = 0$ $y_{iso-C4H10} = 0$ $y_{n-C4H10} = 0$ $y_{ethane} = 0$

We calculate the compressibility factor Z_v of the vapor mixture and fugacity coefficient ϕ_i^V of both components in the vapor phase and the same for the liquid phase. If $Z_L \geq Z_V$ we must change our initial guess for bubble temperature and vapor composition.

Now we calculate the K-values $K_i = \frac{\phi_i^L}{\phi_i^V}$ and solve the Rachford-Rice equation setting $\alpha = 0$ for the bubble point and manipulating the bubble point temperature. The CHE Calculator automatically updates the compressibility factors and fugacity coefficients of both phases.

Next, we recalculate the new values of the vapor composition from $y_i = K_i x_i$

We performed the convergence test. An easy way to do this in the CHE calculator is to click Bubble Point Calculations then click Bubble Point Temperature. Both the original vapor composition and the new vapor composition should match. The final value is $T_{BUBBLE} = -57.34$ in degree Celsius. We can easily convert to Fahrenheit using $1^\circ\text{C} = 33.8^\circ\text{F}$

$$T_{BUBBLE} = -71.2$$

Here is our Excel file for this:

VLE Calculations		Peng-Robinson					
Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O
Molar Composition	1.0000	0.799388	0.099923	0.039969	0.029977	0.029977	0.000765
Temperature, C		82.5500	32.2500	96.6500	134.9500	152.0500	374.1500
Pressure, bar	62.0526						
Saturated Vapor Pressure, bar		72.4962	4.1350	0.4843	0.1083	0.0582	0.0000
Bubble Point Pressure, bar							
Dew Point Pressure, bar							
Saturated Vapor Temperature, C							
Bubble Point Temperature, C	-57.3419						
Dew Point Temperature, C							
K-Values		1.1816	0.4165	0.1926	0.1128	0.0909	0.0734
New K-Values							
Vapor to Feed Molar Ratio (Q=F/V)	0.0000						
The Rachford-Rice Equation	0.0000						
Vapor Phase							
Molar Composition	Mixture						
New Molar Composition		0.944524	0.041616	0.007698	0.003380	0.002726	0.000056
Compressibility Factor		0.844523	0.041617	0.007698	0.003380	0.002726	0.000056
Fugacity Coefficient	0.4687						
Saturated Fugacity Coefficient		0.6827	0.2325	0.0954	0.0474	0.0398	0.0791
Molar Enthalpy, kJ/kmol							
Molar Entropy, kJ/(kmol K)							
Molar Density, kmol/m3							
Molar Heat Capacity, kJ/(kmol K)							
Liquid Phase							
Molar Composition	Mixture						
New Molar Composition		0.799388	0.099923	0.039969	0.029977	0.029977	0.000765
Compressibility Factor	0.2117						
Fugacity Coefficient		0.8066	0.0968	0.0184	0.0053	0.0036	0.0058
Liquid Activity Coefficient							
Molar Enthalpy, kJ/kmol							
Molar Entropy, kJ/(kmol K)							
Molar Density, kmol/m3							
Molar Heat Capacity, kJ/(kmol K)							
Molar Heat of Vaporization, kJ/kmol							
User Data and Calculations: ON							
Bubble point Temperature in F	-71.2153						
Initial K values		0.0190	0.0067	0.0031	0.0018	0.0015	0.0012

Problem B

Problem B: Gibbs Phase Rule and CO₂ Solubility in water

A mass of air comes into contact with cold water and equilibrium is established at a temperature of 20 °C and pressure of 2 bar. Assume that the gas phase follows the Peng-Robinson EOS and the liquid phase is ideal. Use the steam tables to calculate the saturated vapor pressure of water. The molar composition of dry air can be taken as: 78.084 % N₂, 20.946 % O₂, 0.928 % Ar and 0.042 % CO₂

- Calculate the Degrees of Freedom by applying the Gibbs Phase Rule and any other constraints present in the system.
- Calculate the solubility of CO₂ in the water phase. Your answer should be given in mg of CO₂ per liter of pure water.

The Binary Interaction Parameters $\{k_{ij}\}$ for the components in the vapor phase that follows the Peng-Robinson Equation of State can all be set equal to zero

The first step to setting up this problem is to set it up in the CHE calculator. In the data base of the calculator we added all of 5 of our components in this problem (N_2 , O_2 , Ar, CO_2 , and H_2O) and we also set our given binary interaction parameters (BIPS) to zero as stated in the problem, which will be referenced by the calculator in our calculations. We also changed the database from an ideal gas to Peng Robinson because that is our given EOS that we need when making our calculations. An image of the data base set up is shown below:

[illegible]

Once setting up our database, we moved on to completing the degree of freedom analysis (DOF). The equation for calculating the Gibbs phase rule DOF as discussed in the lecture is $DOF = N + 2 - \Pi$. Where N is the number of components in the system, π is the number of phases. In this system we have 5 components (N₂, H₂O, O₂, Ar, CO₂, H₂O) and π is 2 because the system has two phases, liquid and gas. If we plug our values into the equation, we end up with $DOF = 5 + 2 - 2 = 5$.

But our problem statement gives us several constraints that will become the specifications we need to specify our problem enough to solve it. One of the specifications given to us is that the temperature is fixed at 20 degrees Celsius. Another specification is that the problem tells us that the pressure is 2 bar.

At this point, the remaining DOF corresponds to the intensive variables describing the compositions of the two phases. However, several additional constraints remove nearly all these degrees of freedom. First, the liquid phase is assumed to be pure water and ideal, meaning its composition is fully fixed and contributes to no adjustable variables; this eliminates one DOF. Next, the gas-phase composition of the *dry* air is fixed by the problem statement: the molar percentages of N₂, O₂, Ar, and CO₂ are specified and cannot vary independently. Because these four components must retain their given ratios, all gas-phase composition variability except for the water content is removed. This eliminates three more composition degrees of freedom (one for each independent dry-air species beyond the normalization constraint).

After accounting for these restrictions, only **one** meaningful degree of freedom remains: the equilibrium amount of water that transfers into the gas phase. This final DOF corresponds to determining the vapor-phase mole fraction of H₂O that satisfies the phase equilibrium condition

$$f_{H_2O}^{(v)} = f_{H_2O}^{(l)}$$

Which, under the stated assumptions, reduces matching the gas-phase fugacity of water (computed using the Peng–Robinson EOS) with the saturated vapor pressure of water at 20 °C.

Thus, although the formal phase rule yields five degrees of freedom, the specified temperature, pressure, fixed liquid composition, and fixed dry-air composition reduce the problem to a single unknown variable: the water content of the gas phase at equilibrium.

Therefore, our final DOF is 1.

Part two of problem B asks us to calculate CO₂ in the water phase in the units of mg CO₂/ L H₂O. For this part, we needed to use the VLE calculations in the CHE calculator. Since we already added our components to the database, they automatically appear. We input our given temperature and pressure in the fluid conditions section. Since we know the molar composition of dry air, what changes the composition is the addition of liquid water to the gas at equilibrium.

To calculate the new molar compositions with the addition of water we used the same equation that we manipulated in Problem A, which is:

$$Z_i = d_i(1 - Z_{H_2O})$$

Where d_i is the dry composition given to us in the problem statement. This equation will give us the updated composition of each component based on water and because we do not know the molar fraction of water, we need to make an initial guess. For our initial guess of Z_{H_2O} we input 0.01, and made the guess bold and red.

This problem also asks us to calculate the saturated vapor pressure of water. To do that under the fluid conditions in VLE Calculations in the saturated vapor pressure row under the water column, we called the function PSAT. The arguments for the function are the component H2O and the given temperature 20 degrees Celsius.

This gave us a saturated vapor pressure of water that is 0.0238 bar

The next step of this problem requires us to move on to the Vapor phase section of the VLE calculations. In this problem we can assume for the molar compositions that $y_i = z_i$ because in this system, the non-condensable components of air (N_2 , O_2 , Ar, and nearly all CO_2) remain almost entirely in the vapor phase because their solubilities in liquid water at 20 °C are extremely low. As a result, these gases have negligible liquid-phase mole fractions, meaning all of their moles in the system reside in the vapor phase. Therefore, for any component that appears only in the vapor, its overall mole fraction z_i must equal its vapor-phase mole fraction y_i . Only water and CO_2 partitions are between phases, so it is the only species for which we will have x_i .

Due to this fact, we can just make all our vapor phase mole fractions equal to the overall molar compositions. Then we used the vapor phase compositions to calculate the compressibility factor of the mixture. To do this, we called the Zcomp function and the arguments for it are y_i (vapor phase mole fractions) as the molfrac, 20 as temp, 2 as the bar, 1 as our guess for Z. Once we calculated the Z component, we can use it to calculate the fugacity coefficient of each component. To do this we called the PHIC function and inputted the arguments; y_i as molfrac, we called each component in compcol, 20 as temp, 2 as pres and our Z value for the mixture for Zv.

Now that we have the fugacity coefficients of all the components, we can specifically use the fugacity coefficients of CO_2 and H_2O to calculate the fugacity of CO_2 and H_2O . To calculate the fugacity of the two components we used the general equation for fugacity: $f_{CO_2}^v = \varphi_{CO_2} \cdot P \cdot y_{CO_2}$

$$f_{H_2O}^v = \varphi_{H_2O} \cdot P \cdot y_{H_2O}$$

Where f_i^v is the vapor phase fugacity, φ_i is the fugacity coefficient, and P is the pressure which is 2 bar. Now that we have the vapor phase fugacity for CO_2 and H_2O we need to find x_i the

liquid phase molar composition of both components. To find x_{CO_2} we start with the general knowledge that our system is at equilibrium, which means $f^v = f^l$. Now we use Henry's Law which is: $f_{CO_2}^v = y_{CO_2} \cdot H$, where H is Henry's constant that can be calculated using the textbook. Using the fact that vapor and liquid phase are equal, we can set up the equation $f_{CO_2}^v = x_{CO_2} \cdot H$, which then can be turned into $x_{CO_2} = \frac{f_{CO_2}^v}{H}$. That is the equation that will be used to calculate the liquid molar composition of CO2. To do this, we need to calculate Henry's constant for CO2 using the textbook. According to the textbook the equation for Henry's Constant is:

$$\ln H = A + \frac{B}{T} C \ln T + DT, \text{ where H is in bar and T is in Kelvin.}$$

The constants for CO2 are:

$$A = 159.8650745$$

$$B = -8741.55$$

$$C = -21.669$$

$$D = 0.00110259$$

$$T \text{ in K} = 293.15 \text{ K}$$

$$H = e^{\left(159.8650745 - \frac{8741.55}{293.15} - 21.669 \ln(293.15) + 0.00110259(293.15)\right)} = 1442.5570 \text{ bar}$$

Now that we have a value for Henry's constant of CO2 we can solve for the liquid molar composition of CO2:

$$x_{CO_2} = \frac{f_{CO_2}^v}{1442.5570} \text{ and the value given to us is based on our initial guess which we later on changed by goal seeking.}$$

The only other component in the liquid phase is water and to find the molar composition we can use Raoult's law instead of Henry's law because the water is not dilute in the liquid phase like CO2 is. Raoult's law for a liquid that behaves ideally is: $f_i^{(l)} = x_i P_i^{sat}$, and since the system is at equilibrium, we know that $f_i^{(l)} = f_i^{(v)}$, these two equations therefore leave us with the resulting expression $f_i^{(l)} = f_i^{(v)} = x_i P_i^{sat}$. Then we can manipulate this equation to manipulate x and solve the molar liquid composition of water: $x_{H_2O} = \frac{f_{H_2O}^{(v)}}{P_{H_2O}^{sat}}$. We previously calculated the vapor phase fugacity and saturated vapor pressure of water, so we came up with a value for x_{H_2O} .

Once we found values for liquid molar composition, we looked at the mixture column and saw that the mixture was a value greater than 1, which is logically not possible because all of the fractions of the components should equal one whole, so this means that our initial guess for the

molar composition of water (z_{H_2O}) was incorrect because all of our calculations are contingent on that initial guess. To get the correct value we know that liquid phase mixture molar composition must equal 1 and we will achieve this by changing z_{H_2O} , which means we must use goal seek. We used the goal seek numerical methods to change the liquid phase mixture composition cell to 1 by changing the cell for z_{H_2O} . This gave us the correct molar liquid compositions:

$$z_{H_2O} = 0.012078$$

$$x_{H_2O} = 0.999999$$

$$x_{CO_2} = 0.000001$$

Once we had the correct molar we can now solve for the solubility of CO₂ in H₂O using the liquid phase molar compositions and the correct unit conversions (done with the help of chatgpt), shown below:

$$\frac{x_{CO_2}(mol)}{x_{H_2O}(mol)} \times \frac{MW_{CO_2}(g)}{MW_{H_2O}(g)} \times 10^6 = \frac{mg\ CO_2}{L\ H_2O}$$

$$\frac{0.000001\ mol}{0.99999\ mol} \times \frac{44.01\ \frac{g}{mol}}{18.0150\ \frac{g}{mol}} \times 10^6 = 1.394\ \frac{mg\ CO_2}{L\ H_2O}$$

The solubility of CO₂ in H₂O is 1.394 mg CO₂/ L H₂O

Screenshot of VLE Calculations for Part B:

N56

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A		B	C	D	E	F	G	H	I	J	K	L	M	N	
VLE Calculations			Peng-Robinson												
Fluid Conditions		Mixture	H2O	N2	O2	Ar	CO2								
1	Molar Composition	1.0000	0.012078	0.771409	0.206930	0.009176	0.000415								
2	Temperature, C	20.0000													
3	Pressure, bar	2.0000													
4															
5	Saturated Vapor Pressure, bar		0.0238												
6	Bubble Point Pressure, bar														
7	Dew Point Pressure, bar														
8															
9	Saturated Vapor Temperature, C														
10	Bubble Point Temperature, C														
11	Dew Point Temperature, C														
12															
13	K-Values														
14	New K-Values														
15															
16	Vapor to Feed Molar Ratio (v=F)														
17	The Rachford-Rice Equation														
18															
19	Vapor Phase	Mixture													
20	Molar Composition	1.0000	0.012078	0.771409	0.206930	0.009176	0.000415								
21	New Molar Composition														
22	Compressibility Factor	0.9986													
23	Fugacity Coefficient		0.9840	0.9990	0.9980	0.9980	0.9921								
24	Saturated Fugacity Coefficient														
25															
26	Molar Enthalpy, kJ/kmol														
27	Molar Entropy, kJ/(kmol K)														
28	Molar Density, kmol/m3														
29	Molar Heat Capacity, kJ/(kmol K)														
30															
31	Liquid Phase	Mixture													
32	Molar Composition	1.0000	0.999999				0.000001								
33	New Molar Composition														
34	Compressibility Factor														
35	Fugacity Coefficient														
36	Liquid Activity Coefficient														
37															
38	Molar Enthalpy, kJ/kmol														
39	Molar Entropy, kJ/(kmol K)														
40	Molar Density, kmol/m3														
41	Molar Heat Capacity, kJ/(kmol K)														

Fugacity CO2

0.000823301

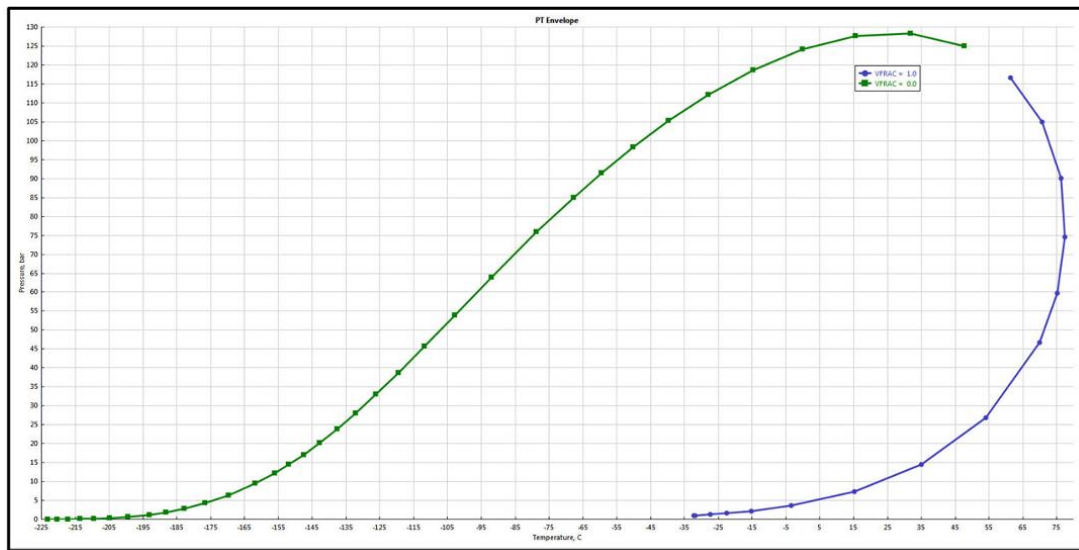
Fugacity H2O

0.023769006

Fugacity CO₂ 0.000823301
Fugacity H₂O 0.023769806

We are asked to calculate the inlet and outlet temperatures to do this. We activate thermodynamics tab in the CHE calculator and enter all the molar compositions (z_i) at the top of the sheet labeled “VLE Calculations”. We also need to put the given inlet pressure, which is 60 bar, in the appropriate cell. The temperature at the inlet of the turbine is equal to the bubble point temperature which we will be solving for. We must first make an appropriate Bubble Point Temp to do this we look at the PT envelope given to us in the problem view graph below:

PT Phase Envelope for the Mixed Refrigerant



At 60 bar the graph reads -90 -degree Celsius. From there we can follow the same steps from part one to solve the bubble point temperature. $T_{BUBBLE} = -94.96$ Which is close to our initial guess.

After finding the inlet temperature, we can find the Entropy and Enthalpy of the liquid phase at the inlet. To do this, we use “enthalpy” and “entropy” functions in the CHE calculator for EOS and enter the arguments: liquid molar composition, pressure, bubble point temperature and

compressibility factor for the liquid phase.

VLE Calculations		Peng-Robinson					
Fluid Conditions	Mixture	N2	CH4	C2H6	C3H8	C4H10	
Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
Temperature, C							
Pressure, bar	60.0000						
Saturated Vapor Pressure, bar		173.2996	31.5444	0.7237	0.0546	0.0053	
Bubble Point Pressure, bar							
Dew Point Pressure, bar							
Saturated Vapor Temperature, C							
Bubble Point Temperature, C	-94.9626						
Dew Point Temperature, C							
K-Values		4.7094	0.9084	0.0687	0.0107	0.0019	
New K-Values							
Vapor to Feed Molar Ratio (α =VF)	0.0000						
The Rachford-Rice Equation	0.0000						
Vapor Phase	Mixture						
Molar Composition	1.0000	0.706408	0.272523	0.020597	0.000000	0.000472	
New Molar Composition	1.0000	0.706407	0.272524	0.020597	0.000000	0.000472	
Compressibility Factor	0.6584						
Fugacity Coefficient		0.8129	0.5401	0.2199	0.1122	0.0535	
Saturated Fugacity Coefficient							
Molar Enthalpy, kJ/kmol							
Molar Entropy, kJ/(kmol K)							
Molar Density, kmol/m ³							
Molar Heat Capacity, kJ/(kmol K)							
Liquid Phase	Mixture						
Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
New Molar Composition							
Compressibility Factor	0.2140						
Fugacity Coefficient		3.8281	0.4906	0.0151	0.0012	0.0001	
Liquid Activity Coefficient							
Molar Enthalpy, kJ/kmol	-98285.6010						
Molar Entropy, kJ/(kmol K)	-258.5379						
Molar Density, kmol/m ³							
Molar Heat Capacity, kJ/(kmol K)							

For the outlet stream, we want the same set up for the molar composition (z_i) but change the pressure to 5 bar. Our unknown for the outlet stream is the vapor fraction, and the outlet temperature. To find the vapor fraction we guess an initial value for the outlet temperature using the PT phase envelope $T = -160^\circ\text{C}$ and we make an initial guess for K_i^0 using $K_i^0 = \frac{p_i^{sat}}{p}$ for supercritical components we use Wilson's P_{sat}

Once the K_i^0 have been computed, we proceed to solve the RR equation to calculate the vapor fraction α

$$\sum \frac{z_i(K_i^0 - 1)}{1 + \alpha(K_i^0 - 1)} = 0$$

We must check that the RR equation is positive when $\alpha = 0$ and negative when $\alpha = 1$ if both aren't true, we must change our initial guess for the K-values.

Once the RR equation have been solved, we can calculate the liquid and vapor composition $x_i = \frac{z_i}{1 + \alpha(K_i^0 - 1)} y_i = x_i K_i^0$

With compositions known both for the vapor and liquid phases, we can calculate the compressibility factor Z_V for the vapor phase mixture, and Z_L for the liquid phase mixture using the CHE Calculator function ZComp. With knowledge of the compressibility factors and compositions we proceed to calculate the fugacity coefficients of each component both in the vapor ϕ_i^V and liquid ϕ_i^L mixtures using the CHE Calculator function PhiC

Now we can calculate new values for the K-values from the equation $K_i^N = \frac{\phi_i^L}{\phi_i^V}$ and perform a convergence test. An easy way to do this in the CHE Calculator is to click “PT Flash Calculations” then click “Calculate vapor fraction”. Doing this sets the original K-values equal to the new K-values, makes the RR equation zero and gives us a value for the vapor fraction. However, this is not the final value.

We must find the outlet entropy and enthalpy values like what we did in the inlet stream, but this time we need both the vapor and liquid components since our vapor fraction is not 0. We also need to use the lever rule $H = \alpha H^V + (1 - \alpha)H^L$, then use the entropy balance to find the correct temperature $S_2 - S_1 = 0$.

As we change the value for temperature, so the entropy balance is satisfied, we must also perform the convergence test for the K-values until the K-values are very close to each other or the same.

Once we have done this our outlet temperature is

$$T_{outlet} = 126.6$$

$$\alpha = 0.28$$

Work done on the turbine is $H_{out} - H_{in}$

$$W = -755 \frac{kJ}{kmol}$$

To convert from kJ/kmol to kJ/kg we can divide by the molar composition of all components by the molecular weight of each component.

$$W = -23.2 \frac{kJ}{kg}$$

Here is our excel file for this:

VLE Calculations		Peng-Robinson					
Fluid Conditions		Mixture	N ₂	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀
Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
Temperature, C	-126.6473						
Pressure, bar	5.0000						
Saturated Vapor Pressure, bar		73.6331	9.0111	0.0813	0.0034	0.0002	
Bubble Point Pressure, bar							
Dew Point Pressure, bar							
Saturated Vapor Temperature, C							
Bubble Point Temperature, C							
Dew Point Temperature, C							
K-Values		37.5026	2.2051	0.0191	0.0006	0.0000	
New K-Values		37.4727	2.2008	0.0190	0.0006	0.0000	
Vapor to Feed Molar Ratio ($\alpha = V/F$)	0.2819	0.0297	0.0043	0.0001	0.0000	0.0000	
The Rachford-Rice Equation	0.0000						
Vapor Phase	Mixture						
Molar Composition	1.0000	0.498286	0.493785	0.007920	0.000000	0.000008	
New Molar Composition							
Compressibility Factor	0.9411						
Fugacity Coefficient		0.9671	0.9226	0.8257	0.7566	0.6889	
Saturated Fugacity Coefficient							
Molar Enthalpy, kJ/kmol	-42463.5675						
Molar Entropy, kJ/(kmol K)	-70.6270						
Molar Density, kmol/m ³							
Molar Heat Capacity, kJ/(kmol K)							
Liquid Phase	Mixture						
Molar Composition	1.0000	0.013287	0.223933	0.414650	0.000000	0.348130	
New Molar Composition							
Compressibility Factor	0.0226						
Fugacity Coefficient		36.2414	2.0305	0.0157	0.0005	0.0000	
Liquid Activity Coefficient							
Molar Enthalpy, kJ/kmol	-121249.1733						
Molar Entropy, kJ/(kmol K)	-332.2989						
Molar Density, kmol/m ³							
Molar Heat Capacity, kJ/(kmol K)							
Molar Heat of Vaporization, kJ/kmol							
User Data and Calculations: ON							
Outlet Entropy	-258.5379						
Outlet Enthalpy	-99040.8248						
Inlet Entropy	-258.5379						
	0.0000						
Inlet Enthalpy	-98286.6010						
Work, kJ/kmol	-755.2238						
Work, kJ/kg	-23.1800						